

Review

Human movement analysis using stereophotogrammetry Part 2: Instrumental errors

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Abstract

This paper reviews the main aspects involved with the management of instrumental errors associated with video-based optoelectronic stereophotogrammetry. Insights on how such errors propagate to kinematic quantities are of great interest in the field of human movement analysis to improve the precision and reliability of measurements. The review focuses on the technical assessment and analytical compensation procedures to cope with instrumental errors. Relevant contributions dealing with intrinsic sources of systematic and random errors, such as the issues concerning camera calibration and filtering and smoothing of marker position data, are presented. Procedures for marker imaged processing, and missing marker recovery are also surveyed. Methods for checking the accuracy and precision of stereophotogrammetric systems are then reviewed. Finally, since the desired outcome of the movement measurements is a reliable estimate of body segment kinematics, state-of-the-art techniques proposed for minimization of error propagation arising from a cluster of external markers are described.

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1. Introduction

This paper is the second of a series dealing with state-of-the-art methods for the analysis of human movement kinematics. The first part of this review series [1] provided the theoretical background for the reconstruction of segmental and joint kinematics. In particular, with respect to the description of segmental kinematics, it was pointed out that the following information is needed:

- position vector and orientation matrix of an arbitrary local frame for each musculo-skeletal model segment, relative to a selected global frame, in each sampled instant of time (${}^G\mathbf{o}_l$, and ${}^G\mathbf{R}_l$ respectively),

- the position vectors of selected particles of the link segments in the relevant local frames.

While the theoretical background presented in the first paper [1] is more general, in the present paper the attention moves from theory to practice, focusing on a representative class of experimental procedures, which provide such information. Particular emphasis will be placed on the potential sources of inaccuracy arising from the motion measurement process while the rigid body assumption for body segments is kept valid.

The experimental set-up consists of a video-based optoelectronic stereophotogrammetric system (OSS), which allows for a non-invasive estimation of the instantaneous position of points in a 3-D measurement volume. It is worth noting that physical markers are not rigidly associated with the bones [2]. Moreover, even in static conditions, reconstructed marker positions are not stationary, due to errors intrinsic in the measuring system [3]. This critically affects

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the accurate estimation of the instantaneous position and orientation (herein after called ‘pose’) of the musculo-skeletal model segments, relative to the selected global frame. The observed relative movements belong to two classes [4]:

- (1) apparent movements, due to both the systematic and random errors with which marker coordinates are reconstructed by the OSS in the global frame;
- (2) real movements, due to the interposition of both passive and active soft tissues between markers and the underlying bone.

The latter movements have only been extensively considered recently and are reviewed in the next paper of this series [2]. The focus here is on the apparent movement that originates from the OSS itself, largely due to the finite precision of technical measurement systems, their technology and architecture. For this reason, particular attention is given to the main sources of photogrammetric errors, considered here to include the error with which instantaneous marker positions are reconstructed in the photogrammetric frame and the propagation of this error to the estimated pose of other frames, whether local or global [1]. To investigate the nature and the properties of instrument-related errors prior to each data collection session, some recommendations will be discussed regarding camera calibration procedures, and filtering and smoothing of marker position data. This paper also presents various methods for spot checking the accuracy and precision of routine data collection and processing proposed in the literature. Finally, the problem of bone pose estimation from a cluster of noisy external markers is addressed and methods aimed at minimizing the pose estimation errors are reported. The accuracy with which local frame pose and relevant joint kinematics can be estimated during movement will be analyzed making reference to data from a paradigmatic experiment conducted at our facility.

2. Optoelectronic stereophotogrammetry

2.1. Measurement technology

Stereophotogrammetric methods, whose applications cover wide-range (as in the earth sciences or remote sensing from space) and close-range (as in anatomy and biomechanics) measurements, are used to reconstruct 3-D landmark coordinates from photographs [5], radiographs [6,7], and video images [8]. Photogrammetry has been developed as a photography-oriented science since Muybridge’s well-known sequence of a horse in motion dating back to 1878 [9], and is now under continuous development with the aid of computer vision, pattern recognition and artificial intelligence techniques.

Video images have several potential advantages over the other techniques in terms of time consumption, cost, and potential image distortion of the development process, so that video-based optoelectronic systems are nowadays the most

popular in movement analysis. These systems are used to track, by means of a system of charged-couple device (CCD) cameras, the 3-D position of a set of fiducial points, constituted from either retroreflective (passive) or light-emitting (active) markers. Analytical close-range photogrammetry then allows the estimation of 3-D position data from digitized, noisy image data, using the geometrical properties of central projection from multi-camera observations [10].

Retroreflective passive markers are used together with infrared stroboscopic illumination produced by an array of light-emitting diodes mounted around the lens of each camera. Recognition of passive markers in the video frames can be performed either via pattern recognition software [11] or by dedicated hardware circuits [12]. Conversely, active markers are pulsed sequentially, so the system can detect automatically each marker by virtue of the pulse timing, and marker tracking is more easily performed. Accuracies and sampling rates of passive marker systems may be not as good as those for the active marker systems [13], although the absence of wires, batteries, and pulsing circuitry on the body of the subject under analysis is an important advantage. The 3-D coordinates of each marker are finally computed based upon the 2-D data from two or more cameras, their known location and internal parameters. More details about the principles underlying marker detection and identification can be found in the recent surveys by Furnée [14], Bontrager [15], and Medved [16].

It should be noted that, although optical sensing is one of the most convenient methods (e.g. compared to magnetic or inertial sensors), it has drawbacks due to marker visibility constraints. These drawbacks are partially overcome by using multiple cameras. In fact, for the reconstruction of 3-D coordinates, each marker must be seen simultaneously by at least two cameras, but in practice more than two are recommended, since markers can become obscured from camera views because of arm swinging, walking aids, subject rotation, etc. [14].

Several sources of inaccuracy affect photogrammetric measurements, resulting in an error on marker coordinates. Instrumental errors are of two types: (1) systematic (instrumental systematic error (ISE)); and (2) random (instrumental random error (IRE)). The former type is in any case associated with a model of the measurement system of limited validity, due either to photogrammetric calibration inaccuracies (bad estimation of model parameters) or to non-linearities that this calibration could not take care of (inadequate model). For instance, a valid camera model should not neglect that optical sensing and subsequent image processing always involve image distortion [17]. The magnitude of the systematic errors depends on the size of the measurement field and on the position that the marker assumes within it [18]. Random errors may be due to electronic noise, marker flickering, i.e. the imprecision with which marker images are converted into image points, the digitizing process itself which transforms marker image

coordinates into their numerical values [3,14], and marker imaged shape distortion which can result from velocity effects, partially obscured marker images, merging of markers with each other or with phantom signals (e.g. strobe illumination of another camera) [14,19].

2.2. Camera calibration methods

Important sources of systematic errors in OSS concern optical distortion occurring in the cameras and their stereo set-up [14]. The major sources of camera distortion include radial distortion due to imperfect lens shape, decentering or misalignment of lens components relative to the optical axis, and thin prism distortions which are generally due to improper lens and camera assembly. The stereo set-up includes the distance between the centers of projection of the cameras, also referred to as the baseline, the field of view and depth of field of the objective and the depth range of the object. The task of calibration, in the first place, is to obtain the best results based on the system at hand. Only when the calibration cannot meet the system specifications may some modification be made to the system design. The objective of calibration is rather the attenuation of the inherent ISE. Dedicated experiments, typically performed by the manufacturers, lead to the design of appropriate correction algorithms, incorporated into the stereo camera calibration procedure proposed to the final user, aimed at estimating both the internal and external parameters of each camera. Internal parameters determine how the image coordinates of a point are derived, given the spatial position of the point with respect to the camera. External parameters characterize the geometrical relation between the camera and the scene, or between different cameras [20]. Camera calibration is hence an essential step in video-based stereophotogrammetry and aims at determining the geometric and optical characteristics of the cameras (internal parameters) and the position and orientation of the camera frame relative to a certain laboratory frame (external parameters) [14] using 2-D image points and the corresponding known 3-D object points. The computed camera parameters can then relate the location of pixels in the image to object points in the 3-D reference frame.

With the increasing need for higher accuracy measurement, camera calibration has attracted several research efforts in the computer vision community and a variety of approaches were made available. The existing calibration procedures can be broadly classified as [20,21]: (1) iterative, involving non-linear very general camera models; (2) non-iterative, involving the closed-form solution of linear model equations; (3) two-step, involving the serial use of closed-form and iterative solutions. Most of these originate from the popular direct linear transformation (DLT) algorithm introduced by Abdel-Aziz and Karara [22] where a 3-D calibration object with a grid of control points in known positions throughout the volume of interest is simultaneously recorded by all cameras. With this approach the search for the camera parameters involves the identification of a cam-

era model whose detail and complexity may change significantly across methods [20–24]. For instance, the original model underlying DLT was later generalized to accommodate also non-linear image distortion [25] even though with some incongruence [26]. Linear methods have the advantage of computational efficiency but suffer from a lack of accuracy and robustness [27]. Non-linear methods, on the other hand, offer a more accurate and robust solution but are computationally intensive and require good initial estimates [20,28,29]. The two-step methods [20,21] offer a smart solution to overcoming the previous limitations: the first step generates an approximate solution using a linear technique, while the second step improves the linear solution using a non-linear iterative procedure. Nevertheless, if the number of control points is too small or the noise level on image coordinates is too high, even these methods may become unstable and the risk of performing local rather than global optimization in the second step can be severe. For this reason, alternative paradigms based on genetic algorithms have been recently explored [28] and the good performance obtained in terms of convergence, accuracy, and robustness seems promising for future developments.

As regards specific contributions stemming from the movement analysis community some authors specifically investigated the effect on the system accuracy of changing the number and the 3-D configuration of the control points used for camera calibration. It was established that the best accuracy is achieved when the control points are large in number and evenly distributed in the calibration volume [30]. Unfortunately, when the volume of interest is large (e.g. larger than 1 m^3) the construction of a suitable calibration object to be used with the DLT becomes prohibitive and other calibration strategies should be investigated [17].

Woltring [23] first introduced in this field the comprehensive analytical self-calibration method and its simplified version known as simultaneous multi-frame analytical calibration (SMAC), a technique based on a planar calibration object, equipped with a dense grid of known control points. In order to identify all camera parameters, it is sufficient to observe (by at least two convergent cameras) the calibration object in a number of different positions with respect to the cameras. The only requirement consists in the invariance of internal camera parameters between exposures. The SMAC technique allows covering larger volumes than the DLT, e.g. $>1\text{ m}^3$, as in gait analysis, while for very large volumes, e.g. in sport settings, analytical self-calibration may be more suitable [17]. Despite its theoretical and practical advantages, the application of SMAC is limited to a camera configuration where the calibration object can be viewed by all cameras [14]. With the internal parameters assumed to be known from previous SMAC calibrations and a rough estimate available of the external parameters of two of the cameras, Sabel [31] reported on a full calibration achieved by means of a single large marker moved around so as to cover a large part of the observation space.

Self-calibration techniques are becoming more and more important in movement analysis laboratories as they avoid the use of complex 3-D calibration structures. However, unless a good initialization and careful distribution of the control points is chosen, bad solutions may easily be found [32]. Recently, the attention moved to the development of calibration procedures exploiting the epipolar constraint between a 3-D point and its 2-D projections on the target of two cameras. This approach, pioneered by Dapena et al. [33] and further developed in the computer vision community [34], allows the estimation of the principal points of the cameras without the use of any additional device and allowed for on-line calibration of the internal and external parameters only by surveying a rigid bar in motion inside the working volume [35,36]. In this way, the calibration procedure was reduced to a relatively simple task to be performed and the corresponding algorithm achieved an accuracy comparable with that guaranteed by calibration algorithms based on control points of known 3-D coordinates: 0.4 mm root mean square error (RMSE) over a volume with a diagonal of 1.5 m [37]. This procedure closely resembles those implemented by many of the more frequently utilized passive marker systems.

2.3. Compensation for random errors: filtering and smoothing of position data

Human movement data commonly have a low-frequency content, with additive, wide-band noise, and may include impulsive events as associated with impacts [38]. Since the late 1970s, a large number of studies have been dedicated to the issue of treating noise in these situations [39–53]. The majority of these studies have investigated the nature and the properties of the instrumental errors, proposing a wide-range of filtering and smoothing techniques. In fact, while accurate estimation of signal derivatives (e.g. velocity, acceleration) is essential for providing decisive information in movement analysis, wide-band, ‘white’ measurement noise has a deleterious effect on estimated derivatives [53,54]. The original methods were characterized by setting the relevant parameters using a trial-and-error approach [39–42]. More recently, automatic methods using optimization procedures have been introduced ensuring a more object-specific analysis of the signals [43–50,55]. Extensive surveys of these techniques have been provided by Wood [51], Gazzani [52], and Woltring [53]. In particular, the former classified the different techniques, in the time and frequency domain, as: (i) graphical methods; (ii) finite difference techniques; (iii) approximations with least-squares polynomials; (iv) spline functions; (v) digital filtering; and (vi) Fourier analysis. Particularly popular among biomechanists is the generalized cross-validation with splines (GCVSPL) approach proposed by Woltring [45,53] where a regularization parameter is determined that provides the best estimation of marker trajectory in the (weighted) least-squares sense [56]. Corresponding software designed ad hoc for smoothing and numerical

differentiation of motion data are also available as freeware packages [45,56].

All these methods consider human motion signals as stationary, which, however, is a non-realistic assumption. For example, in sport activities there is a dramatic change in the frequency content of the overall kinematic signals due to the transition from high to low frequency components over time. Consequently, the choice of a single cut-off frequency cannot provide an unbiased estimation of position and, particularly, of velocity and acceleration. Recently, alternative methods have been proposed to address this problem, which are based on time-scale (e.g. see the review by Chau [57]) or time-frequency [58,59] filtering. In these methods, the differentiation process is split into two cascading steps [54]: (1) an adaptive time/scale-frequency filtering; and (2) a finite-difference calculation of the n th order derivatives of the resulting filtered signal. Results are very encouraging and substantial improvements over traditional techniques were observed.

More recently, efforts to reduce instrument-related noise have moved to the assessment and compensation of experimental artifacts due to soft tissue movement [2], and inaccuracies associated with the identification of anatomical landmarks [60].

2.4. Marker imaged processing

The fundamental aim of every OSS is to make available the 3-D coordinates of selected markers, either active or retroreflective, in a measurement field. With this regard: (1) markers need to be recognized in acquired images; (2) a significant data reduction procedure is involved in the transformation of an imaged marker into its geometric coordinates. Both these steps are potential sources of measurement errors and a short summary is worthwhile.

CCD-based OSSs commonly use electronically shuttered cameras that enhance marker contrast under synchronous strobed illumination [14]. Since markers get brighter than the background, a simple threshold detection algorithm can extract the 2-D coordinates of the pixels covered by a marker. Moreover, selective marker extraction from a background can be further enhanced by a bidimensional cross-correlation template-matching method [12,61]. In this way, the recognition of markers is accomplished by also using information about their shape, which must be spherical.

Once marker contours are obtained these need to be converted into a single coordinate pair per marker. This is the object of the marker center estimation procedure whose objectives are further real-time data reduction and subpixel definition of marker centroid 2-D coordinates which are imaged across several lines of the pixel grid [62]. Marker centroid location can be estimated by different algorithms, providing rather different accuracies [63]. Geometric centroid [61] and circle-fitting [64] are among the methods implemented in current OSSs [16]. The accuracy of geometric

centroid estimation was assessed with a simulation approach [63,65], showing a trade-off of error against marker size. The circle-fitting procedure estimates the marker centroid as the center of the pixelized circle based on a least-squares approach [66]. A further method was developed [63] which gives better results than the previous ones and also does not require the knowledge of the marker image radius to be effective. Unfortunately, its iterative nature limits its use in real-time applications.

2.5. Handling missing markers

Usually, the process of 3-D trajectory reconstruction does not present great difficulties when all markers are visible from at least two cameras. Unfortunately, due to visibility constraints and/or to the inability of the marker recognition technique to distinguish neighboring markers, this process, called ‘tracking’, may be not straightforward and may require user guidance thus limiting real-time applications and increasing post-processing time. This may be a supplementary source of error and should be treated appropriately.

A procedure which includes an anatomically-based human model during the reconstruction and tracking process has recently provided some preliminary results in terms of robustness, and has decreased the need for human intervention in marker tracking [67]. A common inconvenience is related to the handling of missing marker coordinates [68], which may invalidate some experimental trials. The recovery of missing marker coordinates can be performed using the rigid-body assumption. However, the relative movement between the available markers and the underlying bone negatively affects the results [69]. Temporal interpolation proved to be an effective solution in some specific cases [68]. A recent paper by Desjardins et al. [70] proposed an automated procedure based on geometry constraints for retrieving hidden markers without being restricted to the usual rigid-body assumption.

3. The estimate of stereophotogrammetric errors

This discussion will begin with the assumption that camera calibration has been performed successfully and proper filtering and smoothing tools are available. This is a realistic condition for most users who simply follow the directions of the OSS manufacturer. The performance of the system, namely accuracy and precision, may still change due a large number of factors. These include the adequacy and the quality of the system itself, but also parameters related to the specific laboratory set-up, such as the number and location of the cameras, the size of the measurement volume, the size and the shape of the calibration object used, and also the care of the user in performing the calibration procedure. Manufacturers typically report that the accuracy with which marker positions are reconstructed

within a certain field of measurement is about 1:3000 of the diagonal of the calibrated volume. The accuracies reported by Shortis [71] in morphometric studies ranged from 1:5000 with 4 cameras to 1:15000 using 36 cameras. Schmid [72] refers to an achievable accuracy of distance measurements in recent commercially available video-based OSS in the range 0.09–1.77% and higher of the measurement field.

The values reported above are largely acceptable in human movement analysis. Nevertheless, the assessment of actual accuracy and precision of marker position measurements deserves an ad hoc investigation during routine use in the laboratory [3]. In fact, understanding the limitations of the available OSS is essential for proper applications in segmental and joint kinematics estimates [73]. Although the manufacturers provide general figures, actual estimates of these fundamental quantities should be obtained directly by the user prior to every experimental session by using spot-checks, i.e. tests that the user may perform easily for verification of the preservation of the OSS performance [3]. Several of such tests have been proposed in the literature, based on different target measurements.

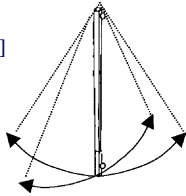
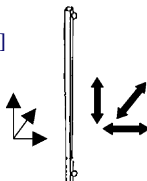
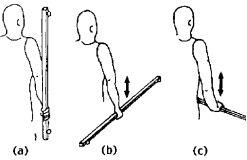
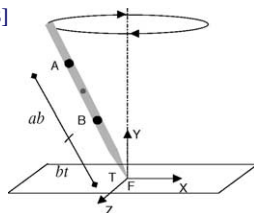
3.1. Inter-marker distance measurement

This group of tests usually involves very simple devices manually driven by an operator within the measurement volume. They are designed so that they can be performed for acceptance testing (yes/no logic) and for validation at the user’s premises [14]. All these procedures aim at revealing the preservation of marker relative distance throughout the measurement volume, assumed to be a good measure of the calibration maintenance. Quantities involved in the performance assessment are statistical figures of merit, mainly expressing precision and accuracy. A classic method to estimate the instrumental error consists of the static or dynamic recording of a rigid bar carrying at least two markers placed at a known distance. The inter-marker distance is then estimated by the OSS in each sampled instant in time and the systematic and/or random components of its deviation from a gold standard measurement are evaluated [3,74–78]. An overview of some dynamic tests proposed in the literature based on inter-marker distance estimation is presented in Table 1.

Della Croce and Cappozzo [3] introduced the movement analysis laboratory (MAL) test aimed at estimating both accuracy and precision in determining landmark trajectories using OSSs. The test also allows distinguishing between different error sources. Markers are stuck on a rod aligned to a tip at one of the extremities. The same extremity is located on a point on the floor of known coordinates. The position of the tip of the rod is reconstructed from the measurement of the marker positions and compared to the known coordinates of the point on the floor. This allows for an estimate of the “static” accuracy. A static precision is also assessed by analyzing the high frequency noise present in the pat-

Table 1

Spot-checks proposed in the literature for assessing OSS performance based on inter-marker distance measurements

Test	Description	Reference	Year
Pendulum test	A rigid pendulum mounting two markers in known positions is suspended to swing in two orthogonal planes. The 3-D coordinates of the markers are recorded for at least two complete oscillations. The test can be performed both near the center and near to the extremities of the calibrated volume.	[74] 	1993
Full volume test	A stiff rod mounts two spherical markers fixed at each end. The vertical axis of the rod is aligned approximately with the vertical axis of the laboratory frame and then moved parallel to each axis, throughout the entire measurement volume, while the speed of the rod movement is kept steady.	[74] 	1993
Walking test	Very similar to the former. The examiner walks so that his feet hit sequentially four marks stuck on the floor, while holding a bar with different orientations, then stops and maintains the upright posture.	[76,77] 	1995, 1997
MAL test	A stiff rod carries two spherical markers (A and B). A target point (T) is taken in a known position with respect to the two markers, coinciding with the tip of the rod. This is placed in a fixed point (F) on the floor within the walk path. Acquisitions are made while the rod is kept stationary (static test), and while the rod is made to rotate around T by moving the other end along a pseudo-circular trajectory or along two composed orthogonal arches. The latter maneuver is performed manually at a speed approximating that of the physical exercise under analysis.	[3] 	2000

terns of the estimated point coordinates due to instrumental noise and flickering (random error, IRE). With a simple man-operated movement of the rod pivoting around the point on the floor, an additional component of noise related to the non-linearities of the calibrated volume (systematic error, ISE) is estimated by analyzing the pattern of the pivot point positions on the floor determined from the measurement of the trajectories of the markers on the rod. In this way, they showed not only how the magnitude of the errors depends on the marker relative distance (Fig. 1) but also provided estimates of the relative contribution of the different sources of error and therefore of their relative contribution to the overall error. Their results were in accordance with the different sensitivities to experimental errors reported by the OSSs [17]. The ISE (Fig. 1c–d) can be much larger than the IRE if only a few cameras are used and/or if they are not optimally distributed around the calibrated volume. These results confirm the appropriateness and usefulness of dividing the photogrammetric error into the additive and uncorrelated components of IRE and ISE. Countermeasure to reduce the two types of errors are different; an appropriate location and number of the cameras used can reduce the ISE substantially.

A significant advancement over the “MAL test” has been recently proposed by Holden et al. [78] to check the OSS performance not only as a stand-alone measurement system but also in combination with force platform (FP) kinetic measurements. This is particularly relevant when, for example, kinetic and kinematic measurements are combined in inverse dynamics for the estimation of joint moments or powers. The testing procedure considers not only the measurement system inaccuracies (for the FP and the OSS taken separately) but also the integration aspects related with time and space synchronization of the devices [78]. This is achieved by the use of a rigid mechanical tool, equipped with five retroreflective markers, that is manually pivoted about the tip on the force plate through a broad range of angles. Four variables are proposed as indicators of measurement errors: the angular discrepancy between the OSS-based and the FP-based estimation of rod orientation; and the differences between the FP-based and the OSS-based estimation of the three coordinates of the center of pressure location. The proposed test does not automatically reveal the specific source of all errors but simply discloses unacceptable trials and directs the experimenter to additional tests.

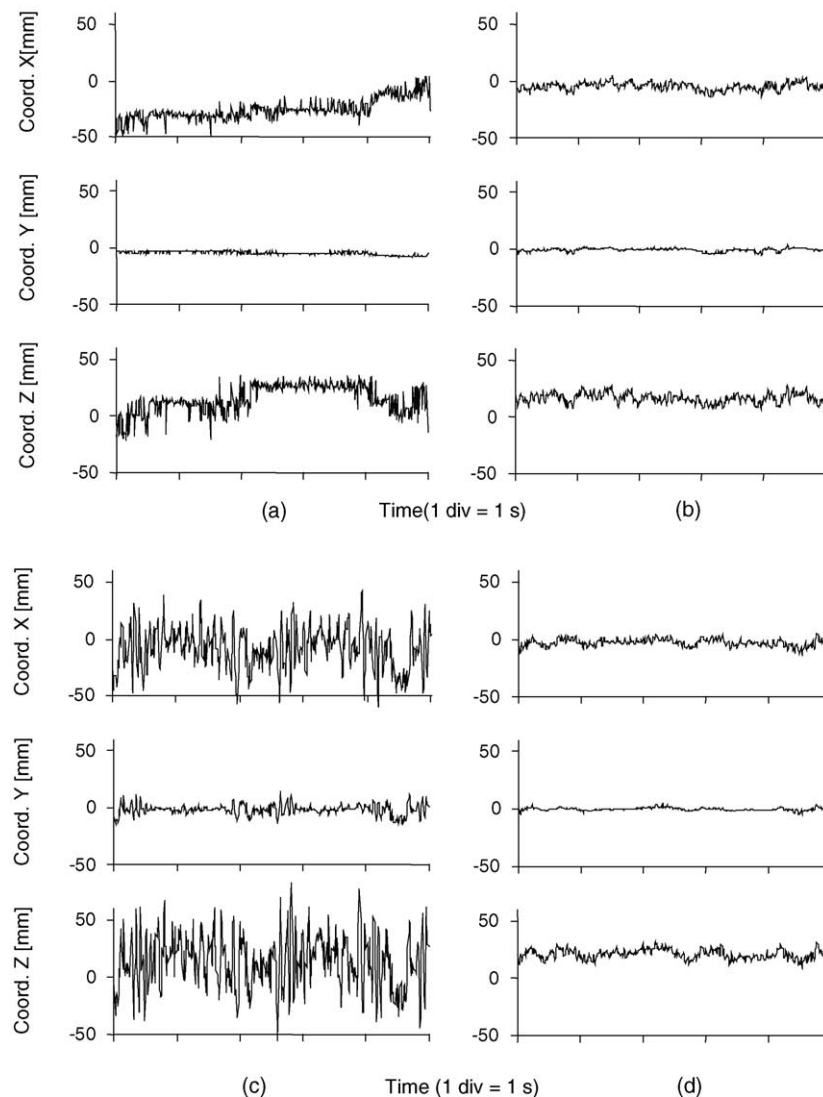


Fig. 1. Representative results from a “MAL test”. Target point coordinates vs. time during a static test (a–b), and during a circular movement (c–d). Two different test-object geometries are considered: near markers, far from the target (a–c) and far markers, close to the target (b–d) (adapted from [3], reproduced by permission of IFMBE).

Specific studies have been devoted to determining the performance of different OSSs that are commercially available [76,77,79–82]. Accuracy and/or precision were reported in a typical gait analysis setting for marker distance estimates. The mean error and accompanying standard deviation ranged from 0.1 (S.D. 0.53 mm) [82] to 5.3 mm (S.D. 4.2 mm) [76]. For most systems examined, both mean error and S.D.s were greater than 1 mm. The absolute error, where reported, ranged from 0.48 [82] to 11.61 mm [77]. The results obtained for several systems are reported in Table 2. Although these tests look simple and the error analysis trivial, the entire procedure is supposed to reflect well some of the conditions in routine clinical movement analysis. Hence, the results are suggested as a compact description of the overall performance of several different OSSs in virtually identical conditions.

3.2. Marker displacement measurement

A number of protocols proposed in the literature investigated marker displacement rather than inter-marker distances. Sometimes this does not require the presence of a human operator but the use of motorized devices.

Some tests use only one marker and take advantage of the knowledge of the trajectories that are expected for the moving object to which the marker is fixed to determine the accuracy of the OSS. Examples of these include the gravity test formerly proposed in the CAMARC framework [74] and the rotating disk test proposed by Cappello et al. [83]. In the latter work, the instantaneous position of a marker firmly attached to the surface of a rotating disk driven by a dc motor was collected and analyzed. The reconstructed trajectories of the marker were found to systematically differ from the

Table 2
Results of the performance tests carried out on different commercially available systems

Author	System	Marker type	Number of cameras (<i>n</i>)	Sampling frequency (Hz)	Test type	Reference length (mm)	Mean error (mm)	Standard deviation (mm)	Mean absolute error (mm)	Maximum error (mm)
Ehara et al. [76]	Peak 5	P	NA	60–2000	D	900	5.3	4.2	5.3	+0.6; –14.2
	Ariel	P	NA	60–400	D	900	5.0	6.0	6.3	+10.8; –26.3
	Vicon 370	P	NA	50, 60, 120, 200	D	900	2.3	1.2	2.3	+4.9; –1.6
	Elite	P	NA	50, 100	D	900	3.2	0.9	3.2	+0.4; –5.6
	Optotrack 3020	A	NA	400/ <i>n</i>	D	900	1.0	0.8	0.9	+0.01; –1.2
	Kinemetrix 3-D	P	NA	50, 100, 200	D	900	3.0	3.8	3.3	+5.1; –12.1
Ehara et al. [77]	Peak 5	P	2	60	D	900	NA	2.04	3.85	+8.10; –10.39
	Ariel APAS	P	2	60	D	900	NA	5.36	11.61	+13.47; –24.07
	Vicon 370	P	6	60	D	900	NA	0.39	0.94	+4.37; –8.57
	Elite Plus	P	4	50	D	900	NA	0.31	0.53	+1.19; –0.96
	Primas	P	2	100	D	900	NA	0.14	1.79	+4.09; –6.14
	Video Locus	P	2	60	D	900	NA	1.45	7.73	+28.23; –7.19
Klein et al. [79]	Ariel	P	2	60	S	500	1.3/0.5	3.5/7.8	NA	NA
Linden et al. [80]	Motion Analysis	P	2	60	D	178.5	0.9–4.4	1.39–3.04	NA	NA
Haggard et al. [81]	Watsmart	A	2	NA	D	100	1.5–2.3	2.1–3.4	NA	NA
Thornton et al. [82]	Kinemetrix 3-D	P	2	NA	S	80	0.1/0.5	0.53/0.63	0.62/0.48	NA

Marker type: P, passive; A, active. Test type: S, static test; D, dynamic test; NA: data not available. Reference length: length of the test object.

expected circular trajectories. This systematic deviation was accounted mainly for by the flickering effect, associated to the fast lightening/darkening of one or more of the camera pixels. The results of the error analysis led to an estimated noise standard deviation between 0.5 and 0.8 mm in the X, Y, and about 2 mm in the Z direction (toward the cameras), which matched well the manufacturer's quoted values. It was concluded that OSS could be used reliably in the quantitative clinical analysis of human tremor, mainly characterized by high frequency and very small segment displacement.

Other studies use two or more markers and determine the accuracy of the system by analyzing their relative motion. In this way, Everaert et al. [84] determined accuracy and precision for an OSS specifically configured for measuring small and slow displacements within a small measurement volume ($0.7\text{ m} \times 0.5\text{ m} \times 0.3\text{ m}$). A reference sliding device was used to control cyclic displacements of two reflective markers over five calibrated ranges, in three different directions and three different zones. Mean and absolute errors were found of 0.034 and 0.094 mm, respectively. The smallest detectable differences in marker positions were about 0.1 mm. This proved that motion analysis systems configured for registration within small volumes may allow measurement of minuscule displacements with great accuracy and may therefore be suitable for many applications in rehabilitation research other than gait analysis.

In another recent paper, Richards [85] proposed a motorized device driving a set of seven markers within the field of measurement, to obtain measurements of the accuracy from several OSSs. This test provides the RMSE of the distance between the markers on the top of a moving bar (as in the "MAL test" [3] and Ehara's performance tests [76,77]), but also the RMSE of the reconstructed angles formed by three markers mounted on a plate located at one extreme of the bar. Moreover, it investigates the ability of the OSS to identify two near markers. Another interesting feature is the ability to perform an analysis on a stationary marker as a function of the minimum distance with an orbiting marker. For most of the OSS assessed, the larger the minimum distance with the rotating marker the better the results. At the extreme condition, where these two markers were only 10 mm apart, all but one of the OSSs that were evaluated misidentified the stationary and variable markers and switched their paths. In addition to these results, the tracking time of the collected data is another parameter used for OSS characterization [85]. The test proposed is an original and robust method for assessing the OSS features, as it provides a complete set of performance measures. However, as is also the case with Ehara's performance tests, this test is probably most suitable for manufacturer use rather than for routine single OSS assessment. The difficulty in manufacturing the device needed for this test is a further limitation to its use for a single system.

4. From marker identification to segment kinematics

A basic objective in movement analysis is the reconstruction of body or bony motion in a photogrammetric frame. We have also pointed out in [1] that the pose in space of a body segment, assumed here to be a rigid body, entails the definition of a local orthogonal frame, namely the technical frame (TF). This is rigid with the segment and is analytically described with respect to a global frame by using a position vector and an orientation matrix [86]. In OSS a reference frame can be readily and non-ambiguously defined starting from the reconstructed coordinates of at least three non-collinear markers [87]. In the light of the results reported so far, it is evident that, due to stereophotogrammetric error on single marker position data, the position vector and orientation matrix of an arbitrary local frame, relative to a selected global frame, will therefore also suffer from a certain degree of inaccuracy. It is our contention that this experimental problem is the major factor that limits the potential scientific and professional contribution of movement analysis. Therefore, in the next sections, we will survey and summarize the experimental and analytical methods aimed at minimizing the propagation of instrumental errors from the kinematics of markers to the kinematics of segmental frames. The aspects of placing external markers such as decisions of where to place them and determining the optimal number per segment, and robust segment pose estimation from noisy marker data will be addressed in sequence.

4.1. Positioning of the external markers

Markers located on the body surface are tracked by the OSS with the ultimate aim of obtaining a reliable estimate of the instantaneous pose of an anatomical frame (AF) [1]. Most protocols so far developed for clinical gait and movement analysis (e.g. see [88–90]), define the AF axes from the location of subcutaneous and/or internal ALs [60]. A set of these frames for lower limb segments has been defined and proposed for standardization [86]. The most common subcutaneous ALs are relatively easy to identify if located by palpation using specific instructions [86]. Within these protocols, such ALs become the suggested location of the set of markers. Of course, the more points on the bone that can be located, the more reliable the definition of the AF will be.

Conversely, there is emerging evidence from the literature that, to obtain a reliable stereophotogrammetric measurement of human movement, different requirements need to be met by the local frame that give some limitations to the AF-based strategy for marker positioning [91]. A few general rules, valid for all protocols, dictated only by experimental requirements and which can facilitate bone pose reconstruction have been provided by Cappozzo et al. [86] and are listed as follows: (a) each marker should be within the field of view of at least two cameras at any given time; (b) markers attached to the same segment should be adequately distributed to minimize position error propagation

to bone orientation; (c) movement between markers and underlying bones should be minimal; (d) it should always be possible to place markers even in the presence of external appliances such as orthoses, prostheses or external fracture fixators; (e) marker disturbance to the subject under analysis should be minimal; (f) marker mounting on the subject should be overall a fast, easy and safe procedure. It was observed that when the cluster of markers is mounted on rigid or semi-rigid fixtures a faster and easier fixation and a better visibility could be obtained [86]. Moreover, a reduction of soft tissue artifacts using these fixtures might be expected [91,92].

It is well evident that ALs often do not satisfactorily comply with the above-mentioned experimental requirements and therefore may not represent ideal locations for marker placement unless using additional fixtures. Hence, a different positioning for the markers is recommended, connected, in the first instance, with TF rather than AF. Consequently, a registration procedure should be provided for the instantaneous reconstruction of the AF pose from that of the estimated TF pose [1] and so an anatomical calibration procedure must be carried out. This can be achieved, e.g. as in the calibrated anatomical system technique (CAST) protocol, without strict rules on the exact placement of markers [86]. Here, the sequence of suggested operations is the following: (i) place the marker on the subjects only according to the general guidelines mentioned above; (ii) define a TF from marker coordinates (see the next paragraph); (iii) by means of a specific static calibration procedure identify the position vectors of selected ALs in the TF; (iv) estimate the time-varying pose of the TF in a global frame during movement; and, finally (v) reconstruct the AF pose from the TF pose with simple vector transformations (see Eq. (2) in [1]).

4.2. The estimation of a segment pose from marker position data

Often, simple non-optimal estimators have been used to calculate the pose of a segment from the coordinates of markers [88,93–96]. With this approach the TF (and similarly the AF) is constructed using simple geometric rules. As an example, Chao and Morrey [87] defined the local frame (x, y, z) upon the knowledge of three non-collinear markers, $\mathbf{m}_1$, $\mathbf{m}_2$, and $\mathbf{m}_3$ by taking $\mathbf{m}_2$ as the origin of the frame, the z -axis set along the line oriented as $\mathbf{m}_1 - \mathbf{m}_2$, the x -axis set perpendicular to the plane defined by the three markers and directed as the vector $\mathbf{v} = (\mathbf{m}_3 - \mathbf{m}_2) \times (\mathbf{m}_1 - \mathbf{m}_2)$, and the y -axis perpendicular to both x and z , that is directed as $(\mathbf{m}_1 - \mathbf{m}_2) \times \mathbf{v}$. Alternatively, a set of three mutually orthogonal axes with the origin in correspondence to the marker cluster centroid can be defined. The axes may be arbitrarily chosen but a particularly significant reference system is that represented by the principal axes of the cluster spatial distribution [4]. All these methods lack consideration of instrumental errors, and assume the cluster of markers to be rigid with the bone and not deformable.

Once the TF has been defined, the position vector and orientation matrix of the TF with respect to the global frame (${}^G\mathbf{o}_1$ and ${}^G\mathbf{R}_1$) are computed by standard matrix algebra. In the simple case cited above, ${}^G\mathbf{o}_1$ is defined by the global coordinates of marker $\mathbf{m}_2$ and ${}^G\mathbf{R}_1$ by the matrix of the unity vectors $[i, j, k]$ of the TF [87]. This procedure assumes that the position of the markers is error-free. When it is not, as in all real circumstances, the errors on marker positions propagate to TF pose and consequently to ${}^G\mathbf{o}_1$ and ${}^G\mathbf{R}_1$ estimates [47]. Furthermore, the redundancy of information related to the availability of three coordinates for each marker of the cluster is not exploited. For this reason, more advanced methods based on least-squares minimization have been proposed to optimally determine segment pose, even in the presence of noise. They have been reported in the literature since the early 1980s and have gained popularity in the last few years. These methods accommodate the deformability of the marker cluster due to photogrammetric noise by exploiting data redundancy. The records associated to a cluster (three times the number of markers) are usually larger than the number of degrees of freedom (DOF) associated with the corresponding TF pose (six); therefore an optimal identification of these DOF, in each instant of time, can be sought.

A short review of these optimization techniques should start with Laub and Shiflett [97] who proposed a method based on linear algebra and matrix perturbation theory. This technique is limited to three non-collinear markers and gives biased estimates [98]. Spoor and Veldpaus [99] first proposed a least-squares method requiring the calculation of the eigenvectors of a 3×3 data matrix. Veldpaus et al. [69] then provided slight improvements to the latter which could avoid computation of the eigenvectors. This method was later refined by the use of the better conditioned singular value decomposition (SVD) algorithm [100], which has the following advantages: (i) an improved stability, essential for ill-conditioned problems; and (ii) the ease of implementation and analysis. The main limitation is that it does not provide a measure of estimation accuracy and cannot explicitly take into account a priori information on noise statistics such as anisotropy. The SVD technique has been extensively presented in the literature [100–102] and found interesting applications also in the case of almost aligned markers [103].

In general terms, the mathematical problem involves the estimation of the orientation matrix ${}^G\mathbf{R}_1$ (or a related orientation vector) and the position vector ${}^G\mathbf{o}_1$ of the TF, on a frame-by-frame basis, starting from the collected coordinates of three or more non-collinear markers associated to the body segment, regarded as affected by additive noise [104]. These markers could be arranged in a rigid or deformable cluster, whose motion can be decomposed into a rigid movement with respect to the bone and a cluster deformation. Considering a cluster of $n \geq 3$ markers, at least three of which not collinear, with relevant 3-D positions in the local and laboratory frame, respectively $\mathbf{x}_k$ and $\mathbf{y}_k$ ($k = 1, 2, \dots, n$), the problem aims at determining the ro-

tation matrix and the translation vector that map optimally the points $\mathbf{x}_k$ to the points $\mathbf{y}_k$. For $\mathbf{x}_k$, a rigid local model is assumed. This can be estimated as the cluster configuration in a given instant of time during the experiment, or an average configuration of the cluster during the analyzed motion [105,106]. The rigid model fitting operator optimally maps the rigid cluster model to the relevant collected noisy configuration. Because of the measurement errors, the mapping is not exact and the residuals can be the object of the minimization. In particular, due to the lack of information about the statistical properties of noise, an unweighted least-squares estimator is well suited. Cappello et al. [50] have proposed a mechanical analogue for this optimization problem with reference to the pose estimation of a deformable cluster on the thigh. Minimizing the sum of the squared distances between ordered pairs of markers corresponds to finding the minimum elastic energy stored in springs, assumed to have the same stiffness value.

To extend the applicability of the former methods, an iterative minimization procedure based on the classical linearization technique has been proposed to take also into account a priori knowledge about noise anisotropy and/or measurement reliability [50]. In this case, body pose is estimated by calculating the translation and rotation that match, in the weighted least-squares sense, the cluster model to the actual cluster as it has been collected by the measurement system. Referring again to the previous mechanical analogue, the different weights that are given both to the markers and to the directions according to known different degrees of reliability can be physically interpreted as different stiffnesses of the springs. The minimization procedure is in this case exploited iteratively and, consequently, it is less efficient than the SVD. It offers, nevertheless, some valuable features such as the capability to adjust to noise conditions and, above all, the possibility to provide useful measures of the accuracy of the estimated DOF. The main limitation is the definition of the weighting matrix since it is difficult to characterize accurately noise components and their mutual relationships.

4.3. Marker cluster design

A further critical aspect of TF pose estimation is the optimal design of the marker clusters, which has not received much attention in the literature. In the only paper [4] known to the authors, the performance of a cluster geometrical configuration was shown to depend on the number of markers (n), their relative geometry, and the position and orientation of the cluster with respect to the target anatomical landmarks. The error propagation to TF pose (here θ defines the orientation vector of the TF) was quantified by the following expressions:

$$\text{rms}||\Delta\mathbf{o}|| = \sigma\sqrt{\frac{3}{n}} \quad \text{rms}||\Delta\theta|| = \frac{\sigma}{r}\sqrt{\frac{\mu}{n}}$$

where $\mu = 4.5$ for a 3-D isotropic marker distribution, $\mu = 5$ for a 2-D isotropic marker distribution, σ the standard deviation of the noise superimposed to the marker coordinates, and r a size index calculated as the rms distance of the markers from their mean position. Interestingly, the study advised that: (a) having the number of markers equal to four seems to be a good compromise between accuracy and practicality; (b) the mean radius of the cluster should be greater than ten times the assessed standard deviation of the experimental error; (c) the three-dimensionality of the cluster is not a critical factor; (d) in quasi-planar clusters, the 2-D isotropic index should be higher than 0.5; and (e) the longest principal axis of the cluster should be oriented toward the relevant AL.

4.4. A paradigmatic experiment

The methods for bone pose reconstruction reviewed so far, and in particular the non-optimal and least-squares techniques, have been compared in terms of their ability in providing a reliable estimate of joint kinematics. Lucchetti et al. [107] provided a good benchmark using reliable information on the target joint kinematics. Experiments were carried out on one patient with a 1 DOF Kotz knee endoprosthesis. The

Table 3

Root mean square errors associated with estimates of knee joint kinematic variables obtained by non-optimal and optimal bone pose reconstruction methods

	Ab/adduction (°)		Anterior displacement (mm)	
	LW	LS	LW	LS
Non-optimal	5.4	4.2	12.4	12.9
Optimal	2.4	2.5	11.9	13.2
	Internal/external rotation (°)		Distraction (mm)	
	LW	LS	LW	LS
Non-optimal	5.7	3.2	7.0	5.3
Optimal	4.1	1.1	6.7	4.2
	Flexion/extension (°)		Lateral displacement (mm)	
	LW	LS	LW	LS
Non-optimal	–	6.7	13.7	11.1
Optimal	–	5.0	13.0	10.1

Results relative to a level walking test at 1.37 m/s (LW) and a lower limb forward swing (LS).

patient wore clusters of four skin markers on the pelvis, and right thigh and shank. Level walking and swinging of the lower limb with the knee hyperextended were performed. The presence of the prosthesis gave the opportunity to assess the validity of the two pose estimation techniques since the estimated knee translation and rotation components, except for flexion-extension during level walking, totally represented errors. The results showed that the optimal approach (four markers) gave some advantages compared to the non-optimal one (three markers) and took some DOF values much closer to the expected real value (Table 3). Nevertheless, even though some improvements were observed mainly in the estimation of the rotational variables, the least-squares approach is not completely satisfactory due to the important residual inaccuracies ranging from 2.4° to 4.1° in the angular displacements and from 4.2 to 13.2 mm in linear displacements. This, in turn, means that the errors that affect the reconstructed marker coordinates have characteristics which go well beyond those that are required for this approach to be effective (normal distribution with zero-mean, uncorrelated).

Similar results have encouraged researchers to consider different techniques where correlated errors are dealt with. In this case, body segments are not simply regarded as rigid segments and soft tissues superimposed are taken into account. Inertial effects, skin sliding and muscle contraction are potential sources of relevant artifacts, which need to be modeled and explicitly compensated for. These new issues will be considered in the next paper of this review series [2].

5. Conclusions

Measurement errors associated with the video-based optoelectronic stereophotogrammetric system itself significantly affect human movement analysis. Instrumental errors may arise at different levels of the measurement chain and are dealt with by several compensation techniques that are used routinely in the movement analysis laboratories. Camera calibration procedures minimize systematic error due to optical distortion phenomena, filtering and smoothing techniques address the issue of random error attenuation, dedicated software takes care of the marker imaged processing and handling of missing markers. An additional critical aspect affecting stereophotogrammetric system performance is the set-up of the laboratory, including the number and placement of cameras, the size of the measurement volume, the outcome of the calibration procedure, and its management over time. Many times a failure to control one of these issues reflects in an unreliable reconstruction of joint kinematics, which is the final objective of the analysis. For this reason, actual estimates of accuracy and precision of the specific system should become a regular practice of laboratories and be considered as a whole with data reporting and interpretation even though manufacturers provide general figures. Several tests of different complexity have been proposed in the litera-

ture, which may contribute the benchmark for the assessment of systematic and random components of instrumental error.

Unfortunately, measurement error on 3-D marker coordinates propagates unpredictably to the estimation of body segment kinematics. Least-squares techniques applied to coordinates of markers allow improvement of the bone pose estimate. These compensation techniques only apply well to experimental conditions when the soft tissues around the bony segments can be disregarded. This is the case where percutaneous or intracortical pins are used which is not usually the case in routine movement analysis. Here, soft tissues play a major role and introduce errors highly correlated with movement. This will need new tools and knowledge to be dealt with and will be considered in the next review in this series [2].

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